alcohol tolerant and produce undesirable compounds, affecting the quality of the wine.

The cultivated wine yeast, *Saccharomyces ellipsoideus*, is used for commercial production. Figure 6.11 shows steps involved in wine production

Infobits

Saccharomyces is called Brewing Yeasts, or Baker's Yeast. The brewing strains can be classified into two groups. The ale strains (*Saccharomyces cerevisiae*) and the lager strains (*Saccharomyces pastorianus* or *Saccharomyces carlsbergenris*). The ale strains are top fermenting strains. Lager strains are hybrid strains of *Saccharomyces cerevisiae* and *Saccharomyces eubayanus* and are often referred to as bottom fermenting. (*Saccharomyces* yeasts can form symbiotic matrices with bacteria and are used to produce Kombucha, Kefir, and Ginger beer).

Steps involved in Wine production

Grapes are stemmed, cleaned and crushed

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Sodium or Potassium Meta – bisulphate is added to check the undesirable microorganism

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Must (crushed grapes) is treated with Sulfur dioxide to kill the wild yeasts and bacteria or sometimes pasteurized to destroy the natural microbiota

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Must is inoculated with *Saccharomyces ellipsoideus* (2.5%) and selected fermentation is carried from 50 to 50000 gallons at 20 to 24°C

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Oak, cement, stone glass lined metal are used as fermentor

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Temperature and time required for fermentation White wine: 10–21°C, 7–12 days; Red wine: 24–27°C, 3–5 days

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In red wine production, after three to five days of fermentation, sufficient tanin and colour is extracted from the pomace and the wine is drawn off for further fermentation

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Racking improves flavour and aroma, where wine is separated from the sediment containing yeast cells as precipitate form

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The wine is subjected to aging at lower temperature. Ageing process is typically much longer for red wine than white wine

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Wines are clarified in a process called fining. Fining is done by filtration through casein, tannin, diatomaceous earth or bentonite clay, asbestos, membrane filters or centrifugation

↓

The wine produced is placed in casks, tank and bottles

After wine production, cork should be used for preventing the entry of air into the bottles. The presence of air allows the growth of vinegar bacteria that convert the ethanol to acetic acid. The final alcohol content of wine varies depending upon the sugar content of the grapes, length of the fermentation and type of strain used.

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